

ISyE 6644OA — Summer 2019 — Test #3 Solutions

(revised 1/20/25)

This test is 120 minutes. You're allowed *three* cheat sheets (6 sides total).

You're allowed the following items:

- Pencil / pen and scratch paper.
- A reasonable calculator.
- *Three* cheat sheets (6 sides total).
- Normal, t , and χ^2 tables, which I will supply. You will not need Kolmogorov-Smirnov or ranking-and-selection tables.

But note that

- You are not allowed to use Arena, even though I'm asking a couple of questions about it.
- This test requires some sort of proctor.
- If you encounter a ProctorTrack issue, contact us immediately (but don't get an ulcer over it).

All questions are 3 points, except #??, which is *1 point*.

1. TRUE or FALSE? If $f(x, y) = cxy$ for all $0 < x < 1$ and $1 < y < 2$, where c is whatever value makes this thing integrate to 1, then X and Y are independent random variables.

Solution: TRUE. (Because $f(x, y) = a(x)b(y)$ factors nicely, and there are no funny limits.) $\square$

2. Show how to generate in Arena a discrete random variable X for which we have

$$\Pr(X = x) = \begin{cases} 0.3 & \text{if } x = -3 \\ 0.6 & \text{if } x = 3.5 \\ 0.1 & \text{if } x = 4 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) `DISC(0.3, -3, 0.6, 3.5, 0.1, 4)`
- (b) `DISC(0.3, -3, 0.9, 3.5, 1.0, 4)`
- (c) `DISC(-3, 0.3, 3.5, 0.9, 4, 1.0)`
- (d) `CONT(0.3, -3, 0.6, 3.5, 0.1, 4)`
- (e) `CONT(0.3, -3, 0.9, 3.5, 1.0, 4)`

Solution: (b). (It's of the form `DISC( $F(x_1), x_1, F(x_2), x_2, \dots$ )`.) $\square$

3. TRUE or FALSE? In our Arena Call Center example, it was possible for entities to be left in the system when it shut down at 7:00 p.m. (even though we stopped allowing customers to enter the system at 6:00 p.m.).

Solution: TRUE — because of the small chance that a callback will occur. $\square$

4. TRUE or FALSE? An entity can be scheduled to visit the *same* resource twice, with *different* service time distributions on the two visits!

Solution: TRUE. $\square$

5. TRUE or FALSE? Arena has a built-in **Input Analyzer** tool that allows for the fitting of certain distributions to data.

Solution: TRUE. $\square$

6. Suppose the continuous random variable X has p.d.f. $f(x) = 2x$ for $0 \leq x \leq 1$. Find the inverse of X 's c.d.f., and thus show how to generate the RV X in terms of a `Unif(0,1)` PRN U .

- (a) $X = U/2$
- (b) $X = 2U$
- (c) $X = U^2$
- (d) $X = \sqrt{U}$
- (e) $X = \sqrt{2U}$

Solution: The c.d.f. is easily shown to be $F(x) = x^2$ for $0 \leq x \leq 1$, so that the Inverse Transform Theorem gives $F(X) = X^2 = U \sim \text{Unif}(0, 1)$. Solving for X , we obtain the desired inverse, $F^{-1}(U) = X = \sqrt{U}$, where we don't worry about the negative square root, since $X \geq 0$. Thus, (d) is the answer. $\square$

7. If U_1 and U_2 are i.i.d. $\text{Unif}(0,1)$ with $U_1 = 0.45$ and $U_2 = 0.45$, use Box-Muller to generate two i.i.d. $\text{Nor}(0,1)$ realizations. (I was lazy and made U_1 and U_2 the same value so that the problem will be easier for me to grade!)

- (a) $Z_1 = 1.2622, Z_2 = -0.0623$
- (b) $Z_1 = 1.2622, Z_2 = 1.2622$
- (c) $Z_1 = -1.2019, Z_2 = 0.3905$
- (d) $Z_1 = -1.2019, Z_2 = -1.2019$
- (e) $Z_1 = 1.2622, Z_2 = -1.2019$

Solution: We have

$$\begin{aligned} Z_1 &= \sqrt{-2\ln(U_1)} \cos(2\pi U_2) = \sqrt{-2\ln(0.45)} \cos(0.9\pi) = -1.2019 \\ Z_2 &= \sqrt{-2\ln(U_1)} \sin(2\pi U_2) = \sqrt{-2\ln(0.45)} \sin(0.9\pi) = 0.3905. \quad \square \end{aligned}$$

Thus, the answer is (c). $\square$

8. Suppose that Z_1, Z_2 , and Z_3 are i.i.d. $\text{Nor}(0,1)$ random variables, and let

$$T = \frac{Z_1}{\sqrt{(Z_2^2 + Z_3^2)/2}}.$$

Find the value of x such that $\Pr(T < x) = 0.99$.

- (a) $x = 4.550$
- (b) $x = 6.965$
- (c) $x = 9.210$
- (d) $x = 11.345$
- (e) $x = -11.345$

Solution: Note that

$$T \sim \frac{\text{Nor}(0, 1)}{\sqrt{\chi^2(2)/2}} \sim t(2).$$

Therefore,

$$0.99 = \Pr(T < x) = \Pr(t(2) < x),$$

so that the quantile $x = t_{0.01,2} = 6.9646$, i.e., answer (b). $\square$

9. Suppose X has the Laplace distribution with p.d.f. $f(x) = \frac{\lambda}{2}e^{-\lambda|x|}$ for $x \in \mathbb{R}$ and $\lambda > 0$. This looks like two exponentials symmetric on both sides of the y -axis. Which of the methods below would be very reasonable to use to generate realizations from this distribution?

- (a) Ask a UGA student
- (b) Box-Muller
- (c) Inverse Transform Method
- (d) Acceptance-Rejection
- (e) Both (c) and (d)

Solution: (a) is a sick joke; (b) is for normal observations; (c) is OK because we can get the c.d.f. in an easy closed form (see the composition notes); and (d) is OK because the p.d.f. is nice and finite. Thus, the correct answer is (e). $\square$

10. Consider a bivariate normal random variable (X, Y) , for which $\mathbf{E}[X] = -3$, $\mathbf{Var}(X) = 4$, $\mathbf{E}[Y] = -2$, $\mathbf{Var}(Y) = 9$, and $\mathbf{Cov}(X, Y) = 2$. Find the Cholesky matrix associated with (X, Y) , i.e., the lower-triangular matrix C such that $\Sigma = CC'$, where Σ is the variance-covariance matrix.

(a) $C = \begin{pmatrix} 2 & 0 \\ 1 & 2\sqrt{2} \end{pmatrix}$

(b) $C = \begin{pmatrix} 2 & 0 \\ 2\sqrt{2} & 1 \end{pmatrix}$

(c) $C = \begin{pmatrix} 4 & 2 \\ 2 & 9 \end{pmatrix}$

(d) $C = \begin{pmatrix} 2 & \sqrt{2} \\ \sqrt{2} & 3 \end{pmatrix}$

Solution: By class notes, we saw that for $k = 2$, we obtain

$$C = \begin{pmatrix} \sqrt{\sigma_{11}} & 0 \\ \frac{\sigma_{12}}{\sqrt{\sigma_{11}}} & \sqrt{\sigma_{22} - \frac{\sigma_{12}^2}{\sigma_{11}}} \end{pmatrix} = \begin{pmatrix} \sqrt{4} & 0 \\ \frac{2}{\sqrt{4}} & \sqrt{9 - \frac{2^2}{4}} \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 1 & 2\sqrt{2} \end{pmatrix}.$$

Thus, the answer is (a). $\square$

11. Consider a nonhomogeneous Poisson arrival process with rate function $\lambda(t) = 2t$ for $t \geq 0$. Find the probability that there will be exactly 2 arrivals between times $t = 1$ and 2.

- (a) 0.056
- (b) 0.112
- (c) 0.224
- (d) 0.448
- (e) 0.896
- (f) None of the above

Solution: The distribution of the number of arrivals between times 1 and 2 is

$$N(2) - N(1) \sim \text{Pois} \left(\int_1^2 \lambda(t) dt \right) \sim \text{Pois} \left(\int_1^2 2t dt \right) \sim \text{Pois}(3).$$

Thus,

$$\Pr(N(2) - N(1) = 2) = \Pr(\text{Pois}(3) = 2) = \frac{e^{-3} 3^2}{2!} = 0.2240. \quad \square$$

This is answer (c). $\square$

12. Suppose we are generating arrivals from a nonhomogeneous Poisson process with rate function $\lambda(t) = 1 + \sin(\pi t)$, so that the maximum rate is $\lambda^* = 2$, which is periodically achieved. Suppose that we generate a *potential* arrival (i.e., one at rate λ^*) at time $t = 0.75$. What is the probability that our usual thinning algorithm will actually accept that potential arrival as an actual arrival? (Note that the π means that calculations are in radians.)

- (a) 0

- (b) 0.146
- (c) 0.5
- (d) 0.854
- (e) 1
- (f) None of the above

Solution: The thinning algorithm says that we accept with probability

$$\frac{\lambda(t)}{\lambda^*} = \frac{1 + \sin(0.75\pi)}{2} = 0.854.$$

This is answer (d). $\square$

13. Suppose $X_1, X_2, \dots$ is an i.i.d. sequence of random variables with mean μ and variance σ^2 . Consider the process $Y_n(t) \equiv \sum_{i=1}^{\lfloor nt \rfloor} (X_i - \mu) / (\sigma\sqrt{n})$ for $t \geq 0$. What is the asymptotic probability that $Y_n(4)$ will be at least 2 as n becomes large?

Hint: Recall that Donsker's Theorem states that $Y_n(t)$ converges to a standard Brownian motion as n becomes large.

- (a) 0
- (b) 0.1587
- (c) 0.5
- (d) 0.8413
- (e) 1
- (f) None of the above

Solution: The Hint reminds us that as $n \rightarrow \infty$, we have

$$Y_n(t) \xrightarrow{\mathcal{D}} \mathcal{W}(t) \sim \text{Nor}(0, t).$$

In particular, $Y_n(4) \xrightarrow{\mathcal{D}} \text{Nor}(0, 4)$, so that

$$\Pr(Y_n(4) \geq 2) \approx \Pr\left(\text{Nor}(0, 1) \geq \frac{2-0}{\sqrt{4}}\right) = \Pr(\text{Nor}(0, 1) \geq 1) = 0.1587.$$

This is choice (b). $\square$

14. Which one of the following properties of a Brownian motion process $\mathcal{W}(t)$ is **FALSE**?

- (a) $\mathcal{W}(0) = 0$.
- (b) $\mathcal{W}(t) \sim \text{Nor}(0, t)$.
- (c) $\mathcal{W}(3) - \mathcal{W}(1)$ has the same distribution as $\mathcal{W}(11) - \mathcal{W}(9)$.
- (d) $\mathcal{W}(3) - \mathcal{W}(1)$ is independent of $\mathcal{W}(4) - \mathcal{W}(2)$.
- (e) $\text{Cov}(\mathcal{W}(3), \mathcal{W}(5)) = 3$.

Solution: (a) and (b) are TRUE because they are fundamental axioms of Brownian motion. (c) is TRUE by stationary increments (another fundamental axiom). (d) is FALSE because it violates independent increments (due to overlapping time intervals). (e) is TRUE since we showed in class that $\text{Cov}(\mathcal{W}(a), \mathcal{W}(b)) = \min(a, b)$.

Thus, the FALSE answer we're looking for is (d). $\square$

15. Find the sample variance of $-10, 10, 0$.

- (a) 0
- (b) 10
- (c) $\sqrt{200}$
- (d) 100
- (e) 200
- (f) None of the above

Solution: $S^2 = 100$. So (d) is the answer. $\square$

16. If $X_1, \dots, X_{10}$ are i.i.d. $\text{Exp}(1/7)$ (i.e., having mean 7), what is the expected value of the sample variance S^2 ?

- (a) $1/49$
- (b) $1/7$
- (c) $10/7$
- (d) 7
- (e) 49

(f) None of the above

Solution: S^2 is always unbiased for the variance of X_i . Thus, we have $E[S^2] = \text{Var}(X_i) = 1/\lambda^2 = 49$, which is answer (e). $\square$

17. TRUE or FALSE? The mean squared error of an estimator is the square of the bias plus the square of its variance.

Solution: FALSE. It's $\text{bias}^2 + \text{var}$. $\square$

18. If $X_1 = 7$, $X_2 = 3$, and $X_3 = 5$ are i.i.d. realizations from a $\text{Nor}(\mu, \sigma^2)$ distribution, what is the value of the maximum likelihood estimate for the variance σ^2 ?

(a) $\sqrt{2.667}$

(b) 2

(c) 2.667

(d) 4

(e) 16

(f) None of the above

Solution: We know from a class example that

$$\hat{\sigma}^2 = \frac{n-1}{n} S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 = 2.667.$$

Thus, the answer is (c). $\square$

19. Suppose that we take three i.i.d. observations $X_1 = 2$, $X_2 = 3$, and $X_3 = 1$ from $X \sim \text{Exp}(\lambda)$. Using the maximum likelihood estimate for λ , find the MLE of $\Pr(X > 2)$.

(a) 0.037

(b) 0.368

(c) 0.5

- (d) 0.632
- (e) 0.963
- (f) None of the above

Solution: By class notes, the MLE of λ is $\hat{\lambda} = 1/\bar{X} = 1/2$. By the Invariance Property of MLEs, we have

$$\widehat{\Pr}(X > 2) = e^{-\hat{\lambda}x} = e^{-(1/2)(2)} = e^{-1} = 0.368.$$

Thus, the answer is (b). $\square$

20. Suppose we're conducting a χ^2 goodness-of-fit test to determine whether or not 100 i.i.d. observations are from a *Johnson* distribution with $s = 4$ unknown parameters a , b , c , and d . (The Johnson distribution is very general and often fits data quite well.) If we divide the observations into $k = 10$ equal-probability intervals and we observe a g-o-f statistic of $\chi_0^2 = 14.2$, will we ACCEPT (i.e., fail to reject) or REJECT the null hypothesis of the Johnson? Use level of significance $\alpha = 0.05$ for your test.

Solution: Note that the χ^2 test has $\nu = k - s - 1 = 10 - 4 - 1 = 5$ degrees of freedom. Then

$$\chi_0^2 = 14.2 > \chi_{0.05,5}^2 = 11.07,$$

so we (easily) REJECT. $\square$

21. TRUE or FALSE? The Kolmogorov-Smirnov test can be used *both* to see (i) if data seem to fit to a particular hypothesized distribution and (ii) if the data are independent.

Solution: FALSE. (It's just a goodness-of-fit test.) $\square$

22. Let's run a simulation whose output is a sequence of consecutive customer waiting times in a crowded store. Which of the following statements is true?
- (a) The waiting times are independent.
 - (b) The waiting times are correlated.

- (c) The waiting times are normally distributed.
- (d) The waiting times are identically distributed throughout the day.

Solution: Just (b). Consecutive waiting times are rarely i.i.d. normal. $\square$

23. Suppose we want to estimate the expected average waiting time (in minutes) for the first $m = 100$ customers at a bank. We make $r = 3$ independent replications of the system, each initialized empty and idle and consisting of 100 waiting times. The resulting replicate means are:

i	1	2	3
Z_i	12	14	11

Find a 95% two-sided confidence interval for the mean average waiting time for the first 100 customers.

- (a) [4.5, 20.1]
- (b) [8.5, 16.1]
- (c) [10.5, 14.1]
- (d) 12.3 ± 5.8
- (e) None of the above.

Solution: The sample mean and sample variance of the 3 replicate means are easily calculated as $\bar{Z}_3 = 12.333$ and $S_Z^2 = 2.333$. For level $\alpha = 0.05$, we have $t_{\alpha/2, r-1} = t_{0.025, 2} = 4.303$, and so the CI is:

$$\begin{aligned}
 \mu &\in \bar{Z}_r \pm t_{\alpha/2, r-1} \sqrt{\frac{S_Z^2}{r}} \\
 &= 12.333 \pm 4.303 \sqrt{\frac{2.333}{3}} \\
 &= 12.333 \pm 3.795 \\
 &= [8.538, 16.128].
 \end{aligned}$$

Thus, the answer is (b). $\square$

24. Suppose that $\mu \in [-30, 90]$ is a 90% confidence interval for the mean cost incurred by a certain inventory policy. Further suppose that this interval was based on 4 independent replications of the underlying inventory system. Unfortunately, the boss has decided that she wants a 95% confidence interval. Can you supply it?

- (a) $[-61.1, 121.1]$
- (b) $[-51.1, 111.1]$
- (c) $[-30, 90]$
- (d) $[-20.5, 99.5]$
- (e) 30 ± 45

Solution: The 90% confidence interval is of the form

$$[-30, 90] = \bar{X} \pm t_{\alpha/2, b-1} y = 30 \pm t_{0.05, 3} y,$$

where the half-length $t_{0.05, 3} y = 60$.

The new 95% CI will therefore be of the form

$$\begin{aligned} \bar{X} \pm t_{0.025, 3} y &= 30 \pm \frac{t_{0.025, 3}}{t_{0.05, 3}} t_{0.05, 3} y \\ &= 30 \pm \frac{3.182}{2.353} 60 \\ &= 30 \pm 81.14 \\ &= [-51.14, 111.14]. \end{aligned}$$

Thus, the answer is (b). $\square$

25. TRUE or FALSE? Welch's method is a graphical technique to estimate truncation (initialization bias) points for steady-state simulation.

Solution: TRUE. $\square$

26. Suppose that we're studying a stochastic process whose covariance function is $R_k = 3 - k$ for $k = 0, \pm 1, \pm 2$, and 0 otherwise. Find the variance of $\bar{X}_3$ (the sample mean of the first 3 observations).

- (a) -0.35

- (b) 0
- (c) 0.65
- (d) 1.48
- (e) 2.11

Solution:

$$\begin{aligned}
 \text{Var}(\bar{X}_n) &= \frac{1}{n} \left[R_0 + 2 \sum_{k=1}^{n-1} \left(1 - \frac{k}{n} \right) R_k \right] \\
 &= \frac{1}{3} \left[3 + 2 \sum_{k=1}^2 \left(1 - \frac{k}{3} \right) (3 - k) \right] \\
 &= \frac{1}{3} \left[3 + 2 \left(\frac{2}{3} \right) 2 + 2 \left(\frac{1}{3} \right) 1 \right] \\
 &= 2.111. \quad \square
 \end{aligned}$$

Thus, the answer is (e). $\square$

27. Consider the following 5 observations:

54 80 75 62 90

If we choose a batch size of 3, calculate all of the overlapping batch means for me.

- (a) 72.2
- (b) 75
- (c) 67.75, 76.75
- (d) 69.7, 72.3, 75.7

Solution:

$$\begin{aligned}
 \bar{X}_{1,3}^o &= \frac{1}{3} \sum_{i=1}^3 X_i = 69.67, \\
 \bar{X}_{2,3}^o &= \frac{1}{3} \sum_{i=2}^4 X_i = 72.33, \text{ and} \\
 \bar{X}_{3,3}^o &= \frac{1}{3} \sum_{i=3}^5 X_i = 75.67.
 \end{aligned}$$

Therefore, the answer is (d). $\square$

28. Which variance reduction method uses the difference $\bar{X} - \bar{Y}$ of two *positively correlated* sample means to get a lower-variance estimator for the difference $\mu_X - \mu_Y$ of the underlying unknown means?
- (a) common random numbers
 - (b) antithetic random numbers
 - (c) stratified sampling
 - (d) composition

Solution: (a). $\square$

29. Which variance reduction method most-closely resembles *regression*, e.g., an estimator for the expected value $E[X]$ of the form $Y = \bar{X} - \beta(C - E[C])$, where $E[C]$ is known and β is a constant, so that Y is unbiased for $E[X]$.
- (a) common random numbers
 - (b) antithetic random numbers
 - (c) control variates
 - (d) composition

Solution: (c). $\square$

30. TRUE or FALSE? If you are using a ranking-and-selection procedure and two competitors happen to fall within the indifference-zone, then you don't really care too much which one you end up selecting.

Solution: TRUE. That's why it's called the IZ! $\square$

31. Suppose we are interested in determining which of 3 soft drinks is most likely to be chosen by a person in a survey. Which type of ranking-and-selection problem is this?
- (a) Normal
 - (b) Bernoulli

- (c) Poisson
- (d) Exponential
- (e) Multinomial

Solution: (e). $\square$

32. TRUE or FALSE? Sequential ranking-and-selection procedures are designed to stop early if one alternative seems to be way out in front of the others.

Solution: TRUE. $\square$

33. Suppose, when designing a ranking-and-selection procedure, you have decided to increase the desired probability of correct selection compared to a previous run of the procedure. What can you expect?

- (a) Sample sizes that are about the same
- (b) Larger sample sizes
- (c) Somewhat lower achieved $\Pr(\text{CS})$
- (d) A larger indifference zone
- (e) Lower confidence in your selection

Solution: (b). $\square$

34. (1 point — this is not a bonus question!) Zombies or Biebers?

Solution: Zoms. $\square$